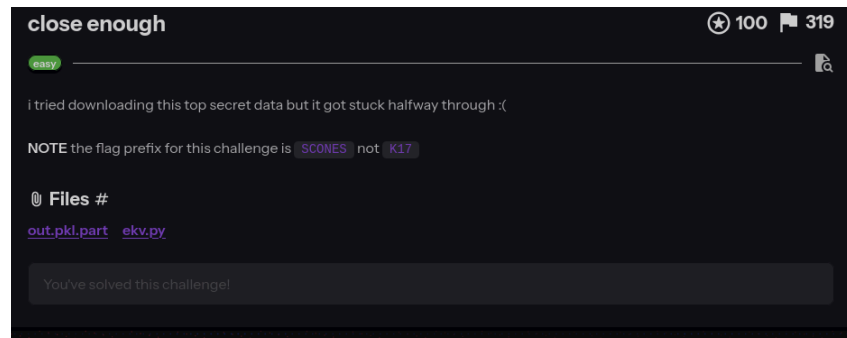


[Forensic] close enough Challenge

Description :

1. "i tried downloading this top secret data but it got stuck halfway through :("
2. the flag prefix for this challenge is SCONES not K17
3. Files: out.pkl.part dan ekv.py

Analyse :



1. We can see that there are 2 files that are given in this challenge. The first thing i do is open the "[ekv.py](#)" files

```
class EncryptedKV:
    def __init__(self, secret):
        self.secret = secret
        self.d = {}

    def __getitem__(self, key):
        num: int = (self.d[key] ^ self.secret)
        return num.to_bytes(-(num.bit_length() // -8)).decode()

    def __setitem__(self, key, value):
        self.d[key] = int.from_bytes(value.encode()) ^ self.secret
```

2. Since the file extension end with .part, it indicates an incomplete download. In Python pickle, a valid stream must end with STOP opcode b'.'.
3. By analyzing ekv.py, the encryption mechanism use a simple XOR operation between the string byte integer and a self.secret key : Ciphertext = Plaintext XOR Secret, because XOR is reversible, we can perform a know plaintext attack using the know flag format to recover the secret key : Secret = Ciphertext XOR Plaintext
4. I created a Python script named exe.py to inspect the dictionary d from out.pkl.part. Since the download was incomplete, the script appends the missing pickle STOP opcode (b'.'.) before unpickling:

```
1  import pickle
2  from ekv import EncryptedKV
3
4  with open("out.pkl.part", "rb") as f:
5      data = f.read()
6
7  if not data.endswith(b'.'.):
8      data += b'.'.
9
10 obj = pickle.loads(data)
11 print(obj.d)
```

5. After i run the solver, i found the flag

```
SyntaxError: invalid syntax
Downloads → python3 exe.py
Decrypted plaintext: K17{a_M0NaD_1s_4_M0n0Id_1n_th3_c4t3g0Ry_of_3Nd0FuNC70r5}
```

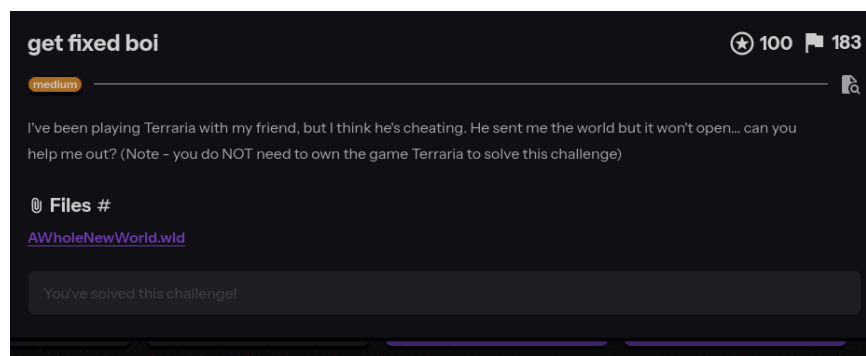
The flag is “K17{a_M0NaD_1s_4_M0n0Id_1n_th3_c4t3g0Ry_of_3Nd0FuNC70r5}”

[Forensic] get fixed boi Challenge

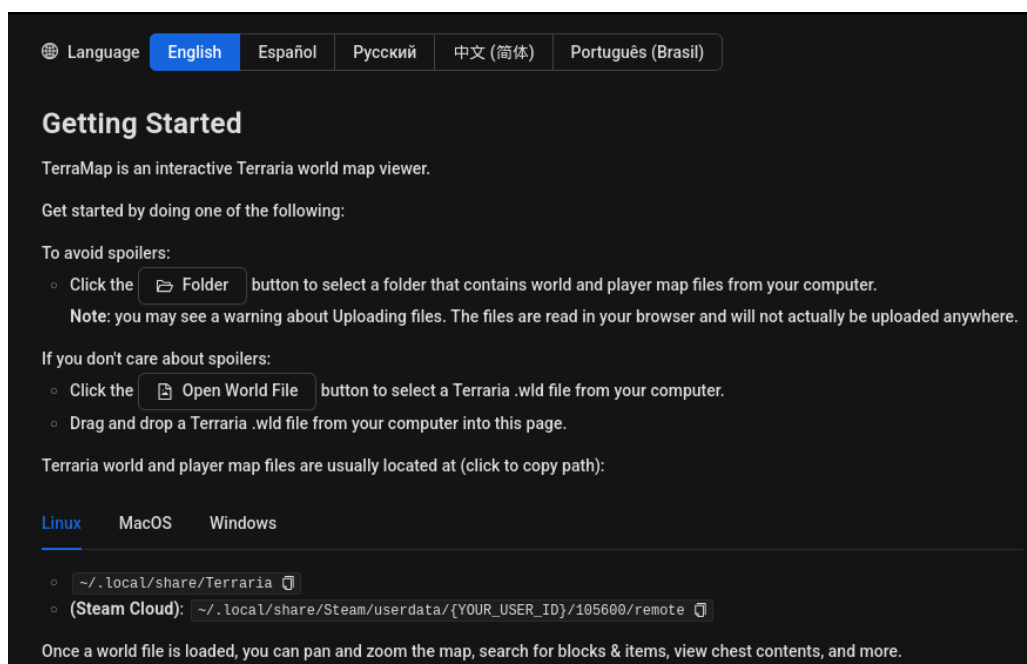
Description :

1. I've been playing Terraria with my friend, but I think he's cheating. He sent me the world but it won't open... can you help me out? (Note - you do NOT need to own the game Terraria to solve this challenge)
2. Files : AWholeNewWorld.wld

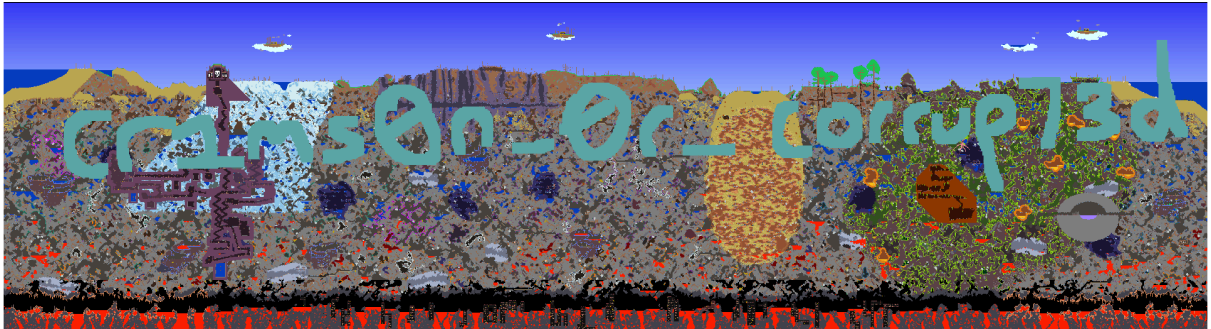
Analyse :



1. Open the online Terraria world map viewer at terramap.github.io in a browser.



2. Click on the Choose File button (or drag and drop) and select the corrupted world file: AWholeNewWorld.wld



3. There is a readable text, {Cr4ms0n_0r_C0rrup73d}

The flag is K17{Cr4ms0n_0r_C0rrup73d}